Introduction to Software Engineering

Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.



Software Engineering Department
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Software Requirements Specification

Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127248	Nguyễn Trần Phương Thúy	Business Analyst	25%	https://github.com/Trananhkhoa2929/ImpactHub/issues/19 https://github.com/Trananhkhoa2929/ImpactHub/issues/16 https://github.com/Trananhkhoa2929/ImpactHub/issues/17 https://github.com/Trananhkhoa2929/ImpactHub/issues/18 https://github.com/Trananhkhoa2929/ImpactHub/issues/26 https://github.com/Trananhkhoa2929/ImpactHub/issues/24 https://github.com/Trananhkhoa2929/ImpactHub/issues/25 https://github.com/Trananhkhoa2929/ImpactHub/issues/22
24127328	Hồng Gia Bảo	Designer	25%	https://github.com/Trananhkhoa2929/ImpactHub/issues/19 https://github.com/Trananhkhoa2929/ImpactHub/issues/20
24127424	Trần Anh Khoa	Project Manager / Tester	25%	https://github.com/Trananhkhoa2929/ImpactHub/issues/23

				https://github.com/Trananhkhoa2929/ImpactHub/issues/24 https://github.com/Trananhkhoa2929/ImpactHub/issues/22
24127508	Lê Nho Duy Phước	Developer	25%	https://github.com/Trananhkhoa2929/ImpactHub/issues/21 https://github.com/Trananhkhoa2929/ImpactHub/issues/20

2 Introduction

2.1 Purpose and Scope

Purpose of this document

This Software Requirements Specification (SRS) describes what ImpactHub must do and the conditions it must meet. It turns the vision, scope and feature priorities agreed in PA0 (Project Idea & Team Contract) and PA1 (Project Proposal & Plan) into numbered, verifiable requirements, use cases and mockups. After the requirements review in Section 10, this version becomes the **requirements baseline** for software design (PA3), test planning (PA4) and implementation (PA5). Any later change to a baselined requirement will be updated in this document and listed in the next PA.

Product scope

ImpactHub is a responsive web application that combines a community forum with fundraising campaigns, simulated donations and donation progress tracking, so that people who want to help can find campaigns approved by an administrator and support them in one place. The scope follows the MoSCoW feature list in PA1 (F01–F13):

Priority	Features	Coverage in this SRS
Must	F01 Account & role management · F02 Community forum · F03 Fundraising campaign management · F04 Campaign approval & moderation · F05 Donation & progress tracking	Fully specified: FR, NFR, business rules, acceptance criteria, use case specifications and mockups

Should	F06 Expense reporting · F07 Email notifications · F08 Volunteer task board	Specified at FR level (FR-06, FR-07, FR-08); implemented if the Must features are finished on time
Could	F09 Payment gateway (sandbox) · F10 Charity marketplace · F11 Google login	Listed for completeness only (F10 as FR-09); not baselined
Won't (this release)	F12 Native mobile app · F13 AI fraud detection & auto-moderation	Out of scope

Out of scope (from PA1 Section 2.4; item 7 is feature F12, Won't):

1. Real-money transactions, payouts or withdrawals to bank accounts. All donations use a demo wallet; any payment gateway runs only in sandbox mode.
2. Official identity verification, legal checks and background checks of users and organizers. Campaign approval is only an internal review by an administrator against the platform rules; it does not determine whether a campaign is legitimate or trustworthy.
3. Quality inspection or authentication of marketplace products, and real shipping or delivery.
4. Real-time push updates such as WebSocket or live chat. Donation progress is updated right after each donation and on page reload.
5. Complex volunteer management such as scheduling, shift planning or location tracking.
6. Advanced financial management such as refunds, invoices, tax receipts or accounting reports.
7. A separate native mobile application.

Intended readers

Reader	How they use this document
Instructor and teaching assistant	Assess the requirements, use case model and prototype against the PA2 rubric
Team lead / project manager	Plan and track PA3–PA5 tasks against the baselined requirements
Business analyst	Maintain the requirements, the traceability matrix and the change list
Designer	Design the UI/UX and the architecture (PA3) from the use cases and mockups
Developers	Implement features that satisfy the FR, NFR and business rules
Tester	Derive acceptance and system test cases (PA4) from the acceptance criteria and use case flows

References

1. T04-PA0 – Project Idea & Team Contract (ImpactHub repository, /pa/PA0).
2. T04-PA1 – Project Proposal & Plan (ImpactHub repository, /pa/PA1; feature list F01–F13, out-of-scope list, technology stack, C4 System Context diagram).
3. Course document "Project Guidelines & Rubric".
4. ImpactHub Figma prototype (links in Section 8).

2.2 Glossary

Term	Definition
ImpactHub	The web application specified in this document.

Guest	A visitor who is not logged in. A guest can browse public forum posts and approved campaigns but cannot post, comment or donate.
Registered user	A person who has an ImpactHub account. Each registered user is assigned one or more of the F01 roles: Donor, Organizer, Volunteer or Admin, and must log in to use the functions of those roles.
Donor	A registered user with the Donor role who makes simulated donations through the demo wallet and follows campaign progress.
Campaign organizer	A registered user with the Organizer role who creates, edits and closes fundraising campaigns, sets donation goals and (Should) posts expense reports and creates volunteer tasks.
Volunteer	A registered user with the Volunteer role who registers for volunteer tasks created by organizers and updates their status (Should, F08).
Administrator (Admin)	A registered user with the Admin role who manages user accounts, reviews and approves or rejects campaigns, moderates reported forum posts and (Should) reviews expense reports.
Person in need (beneficiary)	The individual or group a story or campaign is raised for. They do not need an account.
Story / forum post	Content (text and images) that a registered user publishes in the community forum, e.g. to describe a person in need.
Fundraising campaign	A request for donations for a specific case, with a title, story, cover image, donation goal and deadline, created by a

	campaign organizer. It becomes publicly visible only after an administrator approves it.
Campaign status	The stage of a campaign in its life cycle (e.g. draft, pending approval, approved/active, rejected, closed). The exact states are defined in the business rules (Section 5.3).
Campaign approval	The internal review in which an administrator checks a submitted campaign against the platform rules and approves or rejects it (F04). It is not an official identity, legal or background check and does not determine whether a campaign is legitimate or trustworthy (PA1 Section 2.1).
Moderation	Administrator actions on reported forum posts, such as hiding or removing posts that break the platform rules.
Demo wallet	A virtual balance attached to each registered user account and used only to make simulated donations. It holds no real money and cannot be withdrawn.
Simulated donation	A donation recorded by ImpactHub and deducted from the donor's demo wallet. No real money is transferred.
Donation goal	The target amount a campaign aims to raise.
Amount raised	The total of all successful simulated donations to a campaign.
Donation progress (progress bar)	The amount raised shown as a percentage of the donation goal, displayed as a progress bar on the campaign list and campaign detail pages. It is updated right after each successful donation and on page reload.
Expense report	A record that an organizer publishes on a campaign to show how the raised funds were spent; administrators review it and donors can view it (Should, F06).

Sandbox mode	A payment gateway's test environment (e.g. Stripe or VNPay test mode) in which no real money is charged (Could, F09).
MoSCoW	Prioritisation method: Must have, Should have, Could have, Won't have (this time).
MVP	Minimum viable product: the Must-have features working together end to end.
JWT	JSON Web Token, used to identify a logged-in user in requests to the REST API.
SRS / FR / NFR / BR / UC	Software Requirements Specification / Functional requirement / Non-functional requirement / Business rule / Use case.
RTM	Requirements Traceability Matrix (Section 9), linking requirements to use cases, screens, design elements and test cases.

3 Problem Statement

3.1 Business Context

In Vietnam, help for people in need often starts on social media. A post about a sick child, a family hit by a flood or a student who cannot pay school fees is shared on Facebook, Threads, Zalo or Messenger groups, and readers who want to help transfer money directly to a bank account written in the post. Organizers later report how the money was spent by posting screenshots or photos. Dedicated crowdfunding platforms such as GoFundMe and Give.asia support goal-based fundraising and progress tracking, but they focus on collecting money and do not provide a community space for sharing and discussing stories, or tools to coordinate volunteers and aid distribution.

ImpactHub is a non-commercial, academic project that brings these activities into one web platform: users share stories in a community forum, organizers create campaigns that an administrator reviews and approves before they are published, and donors make simulated donations and follow each campaign's progress.

3.2 Current Problems

ID	Problem	Who is affected	Impact
P1	Stories about people in need are scattered across many social media posts and groups.	Donors, people in need	Cases are hard to find and follow; urgent cases may be missed.

P2	Appeals for money are posted without any review against common rules, so misleading or incomplete appeals are mixed with genuine ones.	Donors, genuine organizers	Donors hesitate to give, and misleading appeals reduce trust in genuine ones.
P3	Social media has no built-in tools for donation goals, collecting donations and showing progress.	Donors, organizers	Donors cannot see how much has been raised or whether the goal has been reached; organizers must track donations by hand.
P4	Reports on how money is spent are informal (screenshots, posts) and hard to find later.	Donors, organizers	Low transparency and accountability after the donation.
P5	Crowdfunding platforms focus on money and do not support community discussion or volunteer coordination.	Organizers, volunteers	Organizers have to combine several tools; volunteers cannot easily find tasks.
P6	Open forums attract spam, misleading or inappropriate posts.	All users, administrators	Without moderation tools, the community becomes less safe and less trusted.

Problem summary (position statement, elaborated from PA1)

The problem of	**stories about people in need being scattered across social media, published without any review and not connected to any fundraising or progress-tracking tool**

affects	donors, campaign organizers, volunteers, and the people in need they want to help
the impact of which is	donors hesitate to give or cannot see how their help is used, organizers struggle to reach donors and show progress, and genuine cases receive less support
a successful solution would	provide one web platform where stories are shared in a moderated forum, campaigns are reviewed and approved by an administrator before they are published, and donors make (simulated) donations and see each campaign's progress toward its goal

3.3 Goals

ID	Goal	Addresses	Related features	Success indicator (checked at PA5 demo)
G1	Provide a single place to share stories and discover campaigns.	P1, P5	F02, F03	A guest can open any approved campaign from the home page through the campaign list.
G2	Publish only campaigns that an administrator has reviewed and approved under the platform rules.	P2	F03, F04	Every campaign visible to the public has been approved by an administrator, and the interface shows the approval status without claiming identity or legal verification.

G3	Make donating simple and progress visible.	P3	F05	A logged-in donor completes a simulated donation from the campaign page, and the new amount raised appears in the progress bar on the next page load.
G4	Keep the community safe through moderation.	P6	F02, F04	An administrator can hide or remove a reported forum post, after which it is no longer visible to other users.
G5	Improve transparency and coordination after fundraising (Should).	P4, P5	F06, F07, F08	Organizers can publish expense reports and volunteer tasks on a campaign.
G6	Deliver all Must-have features as a working, deployed web application within the course schedule and a zero budget.	All	F01–F05	The MVP runs on free-tier hosting by the PA5 deadline (end of week 10).

3.4 Operating Environment

ID	Item	Description
OE-01	Client platform	Responsive web application used in a browser; no installation required. Supported browsers: the two latest versions of Chrome, Edge, Firefox and Safari (HTML5, CSS3, JavaScript ES6+).

OE-02	Screen sizes	Desktop, laptop, tablet and smartphone browsers; reference designs 1440×900 px (desktop) and 360 px (mobile); minimum supported width 360 px.
OE-03	Frontend hosting	Vercel (free tier), deployed from the GitHub repository.
OE-04	Backend and database	Node.js + Express REST API and PostgreSQL database hosted on Render (free tier). A free-tier service may sleep after a period of inactivity, so the first request after a pause can be slow.
OE-05	External services	Cloudinary for storing post and campaign images; an email service for notifications (Should, F07); a payment gateway in sandbox mode (Could, F09).
OE-06	Network	Users need an Internet connection. All traffic between browser and server uses HTTPS.

3.5 Design and Implementation Constraints

ID	Constraint	Source
C-01	The technology stack is React (Vite) + Tailwind CSS for the frontend, Node.js + Express (REST API) for the backend, PostgreSQL for the database, JWT-based authentication (email + password), Cloudinary for images, and Vercel/Render for hosting; GitHub CI, Postman (API tests) and Playwright (frontend tests) for testing.	PA1 Sections 3.2–3.3

C-02	No real money is handled. Donations must use the demo wallet; a payment gateway may only be used in sandbox/test mode.	PA1 Sections 2.1 and 2.4 (F05, F09, out of scope)
C-03	Campaign approval is an internal review by an administrator against the platform rules. The system does not check identity, legal status or background, does not judge whether a campaign is legitimate, and must not present approval as verification.	PA1 Sections 2.1 and 2.4 (F04); WBS 5.7.4
C-04	Updates are not pushed in real time; donation totals and progress are updated right after each donation and on page reload.	PA1 Section 2.4 (F05, out of scope)
C-05	Only free tiers or free allowances of hosting and services are used (infrastructure cost 0 VND); paid upgrades and usage over free limits are excluded.	PA1 Section 4.6
C-06	The team has 4 members and the PA1 estimate is 367 person-hours over the 11-week course, covering only the five Must-have features, documentation, testing and deployment, so the Must-have features take priority.	PA1 Sections 4.3 and 4.6.1
C-07	The project follows the waterfall model with V-model test planning. Requirements are baselined in PA2, design in PA3, test plan in PA4, and implementation and deployment are delivered in PA5 (end of week 10).	Course guidelines; PA1 Section 4.1
C-08	Source code is kept on GitHub; the main branch is protected, every change goes through a pull request reviewed by at least one other member, commit	PA0 Section 3.5; PA1 Section 6

	messages follow Conventional Commits, and every task is tracked on GitHub Projects.	
C-09	The UI follows the team's shared Figma design system (colours, typography, 8-px spacing grid, components).	PA2 design decision
C-10	Only the personal data needed for the features is collected (name, email, password, profile information); passwords are never stored in plain text.	Team decision; password storage is specified in NFR-02

3.6 Assumptions and Dependencies

ID	Assumption / dependency
A-01	Users have an Internet connection and a supported browser (OE-01).
A-02	At least one administrator account exists and an administrator is available to review pending campaigns and reported content.
A-03	Campaign organizers are responsible for the truth of what they submit; administrator approval does not guarantee that a campaign is genuine.
A-04	The free tiers of Vercel, Render and Cloudinary stay available with their current limits until the end of the course.
A-05	Demo data (sample users, posts, campaigns and demo wallet balances) is seeded for testing and the final demo.

4 Stakeholders

#	Stakeholder / user	Description	Main interests
1	Project Team	The team responsible for developing, testing, and maintaining ImpactHub within the approved project scope.	Deliver a functional, maintainable system on schedule and within the agreed scope.
2	Donors	Users who discover and support fundraising campaigns through simulated donations.	Browse approved campaigns, make demo donations, review their donation history and track donation progress.
3	Campaign Organizers	Users who create and manage fundraising campaigns.	Submit and manage campaigns, monitor donations, and follow approval decisions.
4	Administrators	Users responsible for platform administration and moderation.	Manage users, approve or reject campaigns, and moderate reported forum content.
5	Guests	Visitors who have not signed in to ImpactHub.	View publicly available forum posts and approved campaigns; sign up or log in to perform restricted actions.
6	Registered Users	Authenticated users who participate in community features, subject to role-based permissions.	Manage their profiles, create posts, comment, and use functions authorized for their roles.
7	Volunteers	Potential users who may participate in volunteer activities and task coordination if the Should-have Volunteer Task Board (F08) is implemented. Volunteers are a non-Must-have user group; volunteer-specific functionality is not required for the Must-have MVP scope.	Find available tasks, register for them, and update task progress when this optional feature is delivered.

5 Requirements

5.1 Functional Requirements

ID	Requirement	Feature	Priority	Source
FR-01 Account & role management				
FR-01.1	The system shall allow a guest to register an account with an email address and a password.	F01	Must	PA1 §2.4 F01
FR-01.2	The system shall allow a registered user to log in with their email address and password.	F01	Must	PA1 §2.4 F01
FR-01.3	The system shall allow a logged-in user to view and edit their own profile.	F01	Must	PA1 §2.4 F01
FR-01.4	The system shall give each logged-in user access only to the functions of their assigned roles (Donor, Organizer, Volunteer, Admin) and shall deny every other request.	F01	Must	PA1 §2.4 F01
FR-01.5	The system shall allow an administrator to view user accounts and to change the roles assigned to a user.	F01	Must	PA1 §2.1, §2.4 F01
FR-02 Community forum				

FR-02.1	The system shall allow guests and logged-in users to browse public forum posts and to open a post.	F02	Must	PA1 §2.4 F02
FR-02.2	The system shall allow a logged-in user to search forum posts by keyword.	F02	Must	PA1 §2.4 F02
FR-02.3	The system shall allow a logged-in user to create a forum post with text and an optional image.	F02	Must	PA1 §2.4 F02
FR-02.4	The system shall allow a logged-in user to edit and delete their own forum posts.	F02	Must	PA1 WBS 5.5.1
FR-02.5	The system shall allow a logged-in user to comment on and react to a forum post.	F02	Must	PA1 §2.4 F02
FR-02.6	The system shall allow a logged-in user to get a shareable link to a forum post.	F02	Must	PA1 §2.4 F02
FR-03 Fundraising campaign management				
FR-03.1	The system shall allow an organizer to create a fundraising campaign with a title, story, cover image, donation goal and deadline, to save it as a draft and to submit it for administrator review.	F03	Must	PA1 §2.4 F03
FR-03.2	The system shall allow an organizer to edit their own campaigns under the status rules of BR-09.	F03	Must	PA1 §2.4 F03

FR-03.3	The system shall allow an organizer to close their own Approved/Active campaign.	F03	Must	PA1 §2.4 F03
FR-03.4	The system shall allow guests and logged-in users to browse Approved/Active campaigns and to open the detail page of a campaign.	F03	Must	PA1 §2.4 F03
FR-04 Campaign approval & moderation				
FR-04.1	The system shall allow an administrator to list the campaigns in Pending Approval and to review the submitted information of a campaign.	F04	Must	PA1 §2.4 F04
FR-04.2	The system shall allow an administrator to approve or reject a campaign in Pending Approval; a rejection shall require a reason, which is shown to the organizer.	F04	Must	PA1 §2.4 F04
FR-04.3	The system shall make Approved/Active campaigns publicly visible and eligible to receive simulated donations. Closed campaigns shall remain publicly accessible in read-only mode but shall not accept new donations. Draft, Pending Approval, and Rejected campaigns shall not be publicly visible.	F04	Must	PA1 §2.4 F03, F04
FR-04.4	The system shall allow an administrator to list reported forum	F04	Must	PA1 §2.4 F04

	posts and to hide or remove a reported post, or to leave it unchanged.			
FR-05 Donation & progress tracking				
FR-05.1	The system shall allow a logged-in donor to make a simulated donation to an Approved/Active campaign from their demo wallet, without any real-money transfer.	F05	Must	PA1 §2.4 F05
FR-05.2	The system shall show the amount raised, the donation goal and the progress bar of each Approved/Active campaign to guests and logged-in users, updated right after each successful donation and on page reload.	F05	Must	PA1 §2.4 F05
FR-05.3	The system shall allow a logged-in donor to view the history of their successful simulated donations, including date and time, campaign, amount, method and status.	F05	Must	PA1 §2.4 F05
Should-have and Could-have features (FR level only)				
FR-06	The system shall allow organizers to post expense reports for a campaign, administrators to review them and donors to view them.	F06	Should	PA1 §2.4 F06
FR-07	The system shall send emails to users about campaign approval results and donation receipts.	F07	Should	PA1 §2.4 F07

FR-08	The system shall allow organizers to create volunteer tasks and volunteers to register for tasks and update task status (To do, Doing, Done).	F08	Should	PA1 §2.4 F08
FR-09	The system may allow users to list and purchase products using simulated payments, with a specified percentage of each sale allocated to a selected campaign. (Listed only; not baselined.)	F10	Could	PA1 §2.4 F10

5.2 Non-Functional Requirements

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	Under a simulated load of 20 concurrent users, at least 95% of campaign listing and campaign detail requests shall complete within 3 seconds, measured after the back end has been woken up by a warm-up request (OE-04).	Run a JMeter load test with 20 concurrent users and inspect response-time results.

NFR-02	Security	User passwords shall be hashed using bcrypt with a work factor of at least 12. Deployed client-server communication shall use HTTPS. Protected API endpoints shall reject unauthenticated and unauthorized requests.	Inspect password hashing and deployment configuration; execute authentication and role-access API tests.
NFR-03	Usability	At least 4 out of 5 representative users shall complete a forum post creation or simulated donation flow within 3 minutes without assistance.	Conduct usability testing with 5 representative users and record task completion times and success rates.
NFR-04	Scalability	Under a test load of 50 concurrent users, the application shall continue serving campaign listing and detail requests without server errors in at least 99% of requests, measured after the back end has been woken up by a warm-up request (OE-04).	Run a 50-user JMeter test and record request success rates.
NFR-05	Reliability & Availability	During a 24-hour monitoring period in the deployed test environment, the application shall achieve at least 99% availability.	Monitor the deployed application using UptimeRobot and review availability records.
NFR-06	Compatibility	Every page of the Must-have features shall show all content and	Open each page at both widths in the

		controls without horizontal scrolling at viewport widths of 360 px and 1440 px in the two latest versions of Chrome, Edge, Firefox and Safari (OE-01, OE-02).	four browsers and record the result in a checklist.
NFR-07	Accessibility	On every page of the Must-have features, text shall have a contrast ratio of at least 4.5:1, every form field shall have a visible label, and every interactive control shall be reachable and usable with the keyboard and show a visible focus indicator.	Contrast audit of the Figma screens and of the deployed pages; keyboard walkthrough of the main flows.

5.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	Users shall access system features according to their assigned roles: Donor, Organizer, Volunteer, or Admin.	PA1 – F01: Account & Role Management
BR-02	Fundraising campaigns must include a title, story, cover image, donation goal, and deadline.	PA1 – F03: Fundraising Campaign Management
BR-03	Fundraising campaigns must receive administrator approval before becoming publicly visible.	PA1 – F03: Fundraising Campaign Management

BR-04	Only Administrators are authorized to approve or reject fundraising campaigns.	PA1 – F04: Campaign Approval & Moderation
BR-05	Administrators are responsible for moderating reported community forum posts according to platform rules.	PA1 – F04: Campaign Approval & Moderation
BR-06	Campaign approval is limited to the platform's internal review process and does not include official identity verification, legal checks, or background investigations.	PA1 – F04: Campaign Approval & Moderation; Project Scope
BR-07	Donations must be simulated through the demo wallet without processing real-money transactions.	PA1 – F05: Donation & Progress Tracking; Project Constraints
BR-08	Campaign donation totals and progress bars must be updated immediately after each successful simulated donation.	PA1 – F05: Donation & Progress Tracking
BR-09	A campaign may have one of the following statuses: Draft, Pending Approval, Approved/Active, Rejected, or Closed. Draft, Pending Approval, and Rejected campaigns are not publicly visible. Only Approved/Active campaigns can receive simulated donations. An Approved/Active campaign automatically becomes Closed when its deadline is reached. Reaching the donation goal does not automatically close the campaign. An organizer may close an Approved/Active campaign before its deadline	PA1 §2.4 F03–F05; PA2 UC-10, UC-11, UC-13, UC-16

	Closed campaigns remain publicly accessible in read-only mode, with their final donation progress and history preserved. Closed campaigns cannot receive new donations or be edited.	
BR-10	BR-10: Simulated donations are allowed only for Approved/Active campaigns whose deadlines have not passed. Reaching the donation goal does not automatically prevent further donations.	UC-16 Business Rules; PA1 – F03 (Fundraising Campaign Management)
BR-11	Donation Atomicity & Consistency: Failed or interrupted simulated donations must not alter the campaign's total amount raised or progress bar. Financial updates are strictly committed only after a successful transaction verification. A failed donation creates no donation record and no wallet deduction.	PA1 – WBS 5.8.4; UC-16 Exception Flows (E2, E3)
BR-12	Each simulated donation must be at least 10,000 VND and must not exceed the donor's current demo wallet balance.	PA2 prototype (SCR-01_UC-16, SCR-02_UC-16); team decision
BR-13	A donor may add an optional message of up to 200 characters and may donate anonymously. An anonymous donation is shown as "Anonymous" in public donor lists.	PA2 prototype (SCR-01_UC-16, SCR-03_UC-16); team decision
BR-14	A completed simulated donation cannot be cancelled by the donor.	PA2 prototype (SCR-03_UC-16); PA1 §2.4 (refunds are out of scope)

BR-15	Rejecting a campaign requires a reason. The reason is shown to the organizer of the campaign.	PA2 prototype (SCR-02_UC-12, SCR-01_UC-13); team decision
BR-16	A guest may only browse public forum posts and publicly visible campaigns (Approved/Active and Closed), including their donation progress. All other functions require login.	PA1 – F01; Glossary (Guest)
BR-17	A forum post that an administrator has hidden or removed is not shown in public lists, in search results or through its share link.	PA1 – F04; UC-05, UC-06, UC-09, UC-14
BR-18	A user may edit or delete only their own profile and forum posts, an organizer may edit or close only their own campaigns, and a donor may view only their own donation history.	PA1 – WBS 5.6.4; UC-03, UC-11, UC-18
BR-19	During account registration, a password must contain at least 8 characters, including at least one numeric digit. The system must reject passwords that do not meet these conditions and display a validation message.	UC-01, Registration form validation rule (team-defined).

5.4 Acceptance Criteria

Each criterion has a unique identifier and uses Given / When / Then statements. Positive and negative scenarios cover the Must-have features and support PA4 acceptance-test traceability.

F01 – Account & Role Management

AC ID	Given	When	Then

AC-F01-01	A visitor provides valid registration information.	The visitor submits the registration form.	The system creates a user account successfully.
AC-F01-02	A registered user provides valid email and password credentials.	The user submits the login form.	The system authenticates the user and grants access according to the assigned role.
AC-F01-03	An authenticated user accesses their profile.	The user edits and saves profile information.	The system saves and displays the updated information.
AC-F01-04	An email address is already registered.	A visitor attempts to register with that email address.	The system rejects the request and does not create a duplicate account.
AC-F01-05	A user provides incorrect login credentials.	The user submits the login form.	The system denies authentication and displays an error message.
AC-F01-06	A non-administrator attempts to access account administration.	The user requests a restricted management action.	The system denies access and does not disclose restricted account data.
AC-F01-07	Required registration fields are missing or invalid.	The visitor submits the registration form.	The system displays validation errors and does not create an account.
AC-F01-08	An administrator has opened a user account in User Management.	The administrator changes the assigned roles and confirms.	The system saves the new roles, and the user's access follows them.
AC-F01-09	A guest is completing the registration form	The guest submits a password containing fewer	The system rejects the registration, displays an appropriate validation message,

		than 8 characters or no numeric digit	and does not create an account.

F02 – Community Forum

AC ID	Given	When	Then
AC-F02-01	An authenticated user provides post content with an optional image.	The user submits the post.	The system publishes the post in the community forum.
AC-F02-02	An authenticated user views an existing forum post.	The user submits a comment or reaction.	The system records and displays the interaction.
AC-F02-03	A logged-in user enters a keyword in the forum search field.	The user performs a search.	The system displays posts matching the keyword.
AC-F02-04	A user views an existing forum post.	The user selects the share-link function.	The system provides a link to the selected post.
AC-F02-05	A guest is not authenticated.	The guest attempts to publish a post, comment, or react.	The system denies the action and requires authentication.
AC-F02-06	An authenticated user submits an empty or invalid post.	The user attempts to publish the post.	The system shows a validation error and does not publish the post.
AC-F02-07	A forum post has been hidden or removed by an administrator.	A user browses publicly available posts.	The moderated post is not displayed publicly.

F03 – Fundraising Campaign Management

AC ID	Given	When	Then
AC-F03-01	A Campaign Organizer provides a title, story, cover image, donation goal, and deadline.	The organizer submits a new campaign.	The system saves the campaign for administrator review.
AC-F03-02	A Campaign Organizer has a Draft campaign they own.	The organizer edits campaign information and saves it.	The system saves the updated information and keeps the Draft status.
AC-F03-03	A Campaign Organizer has an existing campaign they manage.	The organizer closes the campaign.	The system updates the campaign to Closed status.
AC-F03-04	A campaign has not been approved by an Administrator.	A user browses publicly visible campaigns.	The unapproved campaign is not displayed.
AC-F03-05	Required campaign fields are missing or the donation goal is invalid.	The organizer submits the campaign.	The system rejects the submission and displays validation errors.
AC-F03-06	An organizer does not own or manage a campaign.	The organizer attempts to edit or close that campaign.	The system denies the action and leaves the campaign unchanged.
AC-F03-07	A guest is not authenticated.	The guest attempts to create a fundraising campaign.	The system denies the action and requires authentication.
AC-F03-08	A Campaign Organizer owns an Approved/Active campaign.	The organizer edits a field listed in BR-02 and submits the change.	The system sets the campaign to Pending Approval and hides it from

			the public until an administrator approves it again.
AC-F03-09	A Campaign Organizer has filled in the campaign form.	The organizer selects Save Draft.	The system saves the campaign with Draft status; it is not public and is not sent for review.

F04 – Campaign Approval & Moderation

AC ID	Given	When	Then
AC-F04-01	An Administrator reviews a submitted fundraising campaign.	The Administrator approves the campaign.	The system records approval and makes the campaign publicly visible.
AC-F04-02	An Administrator reviews a submitted fundraising campaign.	The Administrator rejects the campaign and enters a reason.	The system records the rejection and the reason, shows the reason to the organizer and does not publish the campaign.
AC-F04-03	An Administrator reviews a reported forum post.	The Administrator selects Hide post or Remove post.	The post is no longer shown in public lists, in search results or through its share link.
AC-F04-04	A non-administrator attempts to approve or reject a campaign.	The user submits the approval or rejection action.	The system denies the action and preserves the campaign status.

AC-F04-05	A campaign is not in Pending Approval status.	An Administrator attempts to approve or reject it as a pending request.	The system rejects the invalid state transition and preserves the existing status.
AC-F04-06	A non-administrator attempts to moderate a reported forum post.	The user submits a moderation action.	The system denies the action and leaves the post unchanged.
AC-F04-07	An Administrator has selected Reject for a pending campaign.	The Administrator confirms without entering a reason.	The system asks for the reason and leaves the campaign in Pending Approval.

F05 – Donation & Progress Tracking

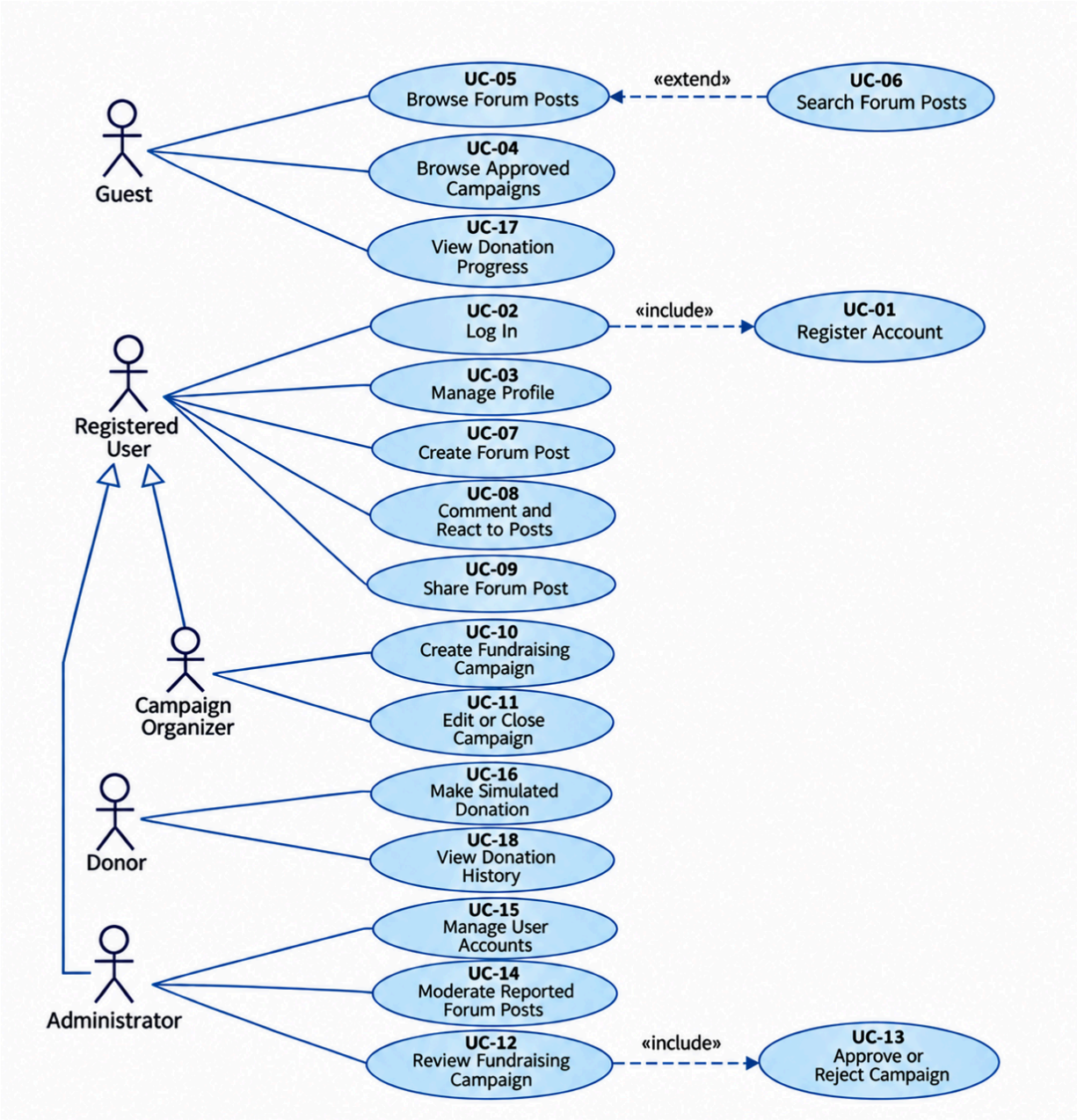
AC ID	Given	When	Then
AC-F05-01	An authenticated donor selects an approved active campaign and provides a valid simulated donation amount through the demo wallet.	The donor confirms the simulated donation.	The system records the simulated donation without transferring real money.
AC-F05-02	A campaign has an existing donation total and fundraising progress bar.	A simulated donation is successfully recorded.	The system updates the campaign total and progress bar.
AC-F05-03	A donation amount is below 10,000 VND or greater	The donor tries to continue.	The system rejects the donation without changing

	than the available demo-wallet balance.		the wallet balance or campaign total.
AC-F05-04	A campaign is pending approval, rejected, or closed.	A donor attempts to make a simulated donation.	The system rejects the donation and does not change campaign progress.
AC-F05-05	A simulated donation fails during processing.	The system handles the failed operation.	The system leaves the wallet balance, donation record, and campaign progress unchanged and displays an error.
AC-F05-06	A guest is not authenticated.	The guest attempts to make a simulated donation.	The system denies the action and requires authentication.
AC-F05-07	A donor has at least one successful simulated donation.	The donor opens My donations.	The system shows the total donated, the number of campaigns supported and the number of donations, and lists each donation with date and time, campaign, amount, method and status.
AC-F05-08	A donor has a failed donation attempt, and other users have made donations.	The donor opens My donations.	The failed attempt and the other users' donations are not listed.
AC-F05-09	A donor has selected "Donate anonymously" on the donation form.	The simulated donation is successfully recorded.	The public donor list shows "Anonymous"

			instead of the donor's name.

6 Use Case Model

6.1 Use Case Diagram



6.2 Actors

Actor	Description
Guest	A visitor who has not logged in. Guests can browse public forum posts and approved fundraising campaigns, but cannot create posts, comment, or make donations.
Registered User	A user who has successfully registered and logged in to ImpactHub. Registered users can manage their profiles, browse forum posts, search content, create posts, comment, react, and share posts.
Donor	A registered user who makes simulated donations through the demo wallet, views their donation history and views fundraising campaign progress.
Campaign Organizer	A registered user who creates, edits, and closes fundraising campaigns, sets donation goals, and manages campaign information.
Administrator	A staff user who reviews submitted fundraising campaigns against internal platform rules, approves or rejects pending campaigns, moderates reported forum posts, and manages user accounts. This does not involve official identity verification, legal verification, or background checks.
Cloudinary	External image storage service (OE-05). It stores the images uploaded with forum posts and campaigns. Secondary actor of UC-07, UC-10 and UC-11.

Actor relationships:

- Donor, Campaign Organizer and Administrator are specialized roles of Registered User.
- A Registered User must log in before accessing authenticated functions.
- Guest can access publicly available information without authentication.
- Administrator has special permissions for campaign approval, content moderation, and account management.
- A Registered User can do everything a Guest can do: browse public forum posts, approved campaigns and donation progress.

6.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR

UC-01	Register Account	Guest	Must	FR-01.1
UC-02	Log In	Registered User	Must	FR-01.2, FR-01.4
UC-03	Manage Profile	Registered User	Must	FR-01.3
UC-04	Browse Approved Campaigns	Guest	Must	FR-03.4, FR-04.3, FR-05.2
UC-05	Browse Forum Posts	Guest	Must	FR-02.1
UC-06	Search Forum Posts	Registered User	Must	FR-02.2
UC-07	Create Forum Post	Registered User	Must	FR-02.3
UC-08	Comment and React to Posts	Registered User	Must	FR-02.5
UC-09	Share Forum Post	Registered User	Must	FR-02.6
UC-10	Create Fundraising Campaign	Campaign Organizer	Must	FR-03.1
UC-11	Edit or Close Campaign	Campaign Organizer	Must	FR-03.2, FR-03.3
UC-12	Review Fundraising Campaign	Administrator	Must	FR-04.1
UC-13	Approve or Reject Campaign	Administrator	Must	FR-04.2, FR-04.3
UC-14	Moderate Reported Forum Posts	Administrator	Must	FR-04.4
UC-15	Manage User Accounts	Administrator	Must	FR-01.5
UC-16	Make Simulated Donation	Donor	Must	FR-05.1, FR-05.2
UC-17	View Donation Progress	Guest	Must	FR-05.2
UC-18	View Donation History	Donor	Must	FR-05.3

7 Use Case Specifications

7.1 Use Case UC-01: Register Account

Use case ID	UC-01
Use case name	Register Account
Brief description	Allows a guest to create a new ImpactHub account.
Primary actor	Guest
Secondary actors	None
Priority	Must
Related requirements	FR-01.1
Trigger	The guest selects the Register option.
Pre-conditions	The guest is not logged in and can access the registration page.
Post-conditions	A new user account is created and stored in the system.
Main flow	1. The guest opens the registration page. 2. The system displays the registration form. 3. The guest enters the required registration information, including email and password. 4. The guest submits the form. 5. The system validates the submitted information. 6. The system creates a new user account. 7. The system displays a registration success message.
Alternative flows	A1 (at step 5): The email address is already registered. The system informs the guest and requests another email address; the flow returns to step 3.
Exception flows	E1 (at step 5): Required information is missing or invalid. The system displays validation messages and requests corrections. E2 (at step 6): An unexpected error occurs. The system displays an error and does not create a duplicate account. E3 – Invalid password (at password validation step): If the password contains fewer than 8 characters or does not include at least one numeric digit, the system rejects the registration attempt, displays a validation message, and allows the user to correct the password. No account is created.

Business rules	BR-01: Users access system features according to their assigned roles.
Non-functional constraints	Passwords must be securely hashed using bcrypt with a work factor of at least 12. The registration form must provide clear labels and validation messages.

7.2 Use Case UC-02: Log In

Use case ID	UC-02
Use case name	Log In
Brief description	Allows a registered user to authenticate and access ImpactHub.
Primary actor	Registered User
Secondary actors	None
Priority	Must
Related requirements	FR-01.2, FR-01.4
Trigger	The user selects the Log In option.
Pre-conditions	The user has an existing account and is not currently logged in.
Post-conditions	The user is authenticated and granted access according to the assigned role.
Main flow	1. The user opens the login page. 2. The system displays the login form. 3. The user enters their email and password. 4. The user submits the form. 5. The system validates the credentials. 6. The system authenticates the user. 7. The system grants access according to the user's assigned role. 8. The system redirects the user to the appropriate page.
Alternative flows	A1 (at step 8): The user requested a protected page before logging in. The system redirects the user to that page after successful authentication, if permitted.
Exception flows	E1 (at step 5): The credentials are incorrect. The system displays an error and allows another attempt. E2 (at step 4): Required login information is missing. The system requests the missing information.
Business rules	BR-01: Users access system features according to their assigned roles.
Non-functional constraints	Passwords must be securely hashed and data transmission must use HTTPS, as specified in NFR-02. The login interface must provide clear error messages.

7.3 Use Case UC-03: Manage Profile

Use case ID	UC-03
Use case name	Manage Profile
Brief description	Allows a registered user to view and update profile information.
Primary actor	Registered User
Secondary actors	None
Priority	Must
Related requirements	FR-01.3
Trigger	The user selects the Profile option.
Pre-conditions	The user is logged in.
Post-conditions	Updated profile information is saved and displayed.
Main flow	1. The user opens their profile page. 2. The system displays the current profile information. 3. The user selects Edit Profile. 4. The system displays editable profile fields. 5. The user updates the desired information. 6. The user selects Save. 7. The system validates the information. 8. The system saves the changes. 9. The system displays the updated profile.
Alternative flows	A1 (at step 5): The user makes no changes and returns to the profile page. The system leaves the existing information unchanged.
Exception flows	E1 (at step 7): The information is invalid or incomplete. The system displays validation messages. E2 (at step 8): The system cannot save the changes. It displays an error and retains the previously saved information.
Business rules	BR-01: Users access system features according to their assigned roles. BR-18: A user may view and edit only their own profile.
Non-functional constraints	Profile information must only be accessible to authorized users. The interface must provide clear field labels and validation feedback.

7.4 Use Case UC-04: Browse Approved Campaigns

Use case ID	UC-04
Use case name	Browse Approved Campaigns
Brief description	Allows guests to view fundraising campaigns approved by an administrator.
Primary actor	Guest
Secondary actors	None

Priority	Must
Related requirements	FR-03.4, FR-04.3, FR-05.2
Trigger	The guest opens the fundraising campaign page.
Pre-conditions	The ImpactHub website is accessible.
Post-conditions	Approved campaigns are displayed; no campaign information is modified.
Main flow	1. The guest opens the campaign page. 2. The system retrieves campaigns with approved status. 3. The system displays the approved campaigns. 4. The guest selects a campaign. 5. The system displays the campaign title, story, donation goal, deadline and donation progress.
Alternative flows	A1 (at step 3): No approved campaigns are available. The system displays an appropriate empty-state message.
Exception flows	E1 (at step 2): The system cannot retrieve campaign information. It displays an error and allows the guest to try again.
Business rules	BR-02: Campaigns must include a title, story, cover image, donation goal and deadline. BR-03: Campaigns must be approved before becoming publicly visible.
Non-functional constraints	Campaign information must be displayed clearly and consistently. The campaign page must meet applicable performance requirements in NFR-01.

7.5 Use Case UC-05: Browse Forum Posts

Use case ID	UC-05
Use case name	Browse Forum Posts
Brief description	Allows guests to view publicly available posts and stories in the community forum.
Primary actor	Guest
Secondary actors	None
Priority	Must
Related requirements	FR-02.1
Trigger	The guest opens the community forum.
Pre-conditions	The ImpactHub website is accessible.
Post-conditions	Public forum posts are displayed; no post is modified.
Main flow	1. The guest opens the community forum page. 2. The system retrieves publicly available forum posts. 3. The system displays the posts. 4. The guest selects a post. 5. The system displays the selected post's content and available information.

Alternative flows	A1 (at step 3): No public posts are available. The system displays an empty-state message.
Exception flows	E1 (at step 2): The system cannot retrieve forum posts. It displays an error message.
Business rules	BR-16: A guest may only browse public forum posts and approved campaigns; other functions require login. BR-17: Posts hidden or removed by an administrator are not shown publicly.
Non-functional constraints	Forum content must be readable and clearly organized. The forum page must meet NFR-06 (compatibility) and NFR-07 (accessibility).

7.6 Use Case UC-06: Search Forum Posts

Use case ID	UC-06
Use case name	Search Forum Posts
Brief description	Allows a registered user to search forum posts using keywords.
Primary actor	Registered User
Secondary actors	None
Priority	Must
Related requirements	FR-02.2
Trigger	The user enters a keyword in the forum search field.
Pre-conditions	The user is logged in and the forum is accessible.
Post-conditions	Matching posts are displayed; no post is modified.
Main flow	1. The user opens the community forum. 2. The system displays the search field. 3. The user enters a keyword. 4. The user submits the search. 5. The system searches available forum posts. 6. The system displays posts matching the keyword.
Alternative flows	A1 (at step 6): No matching posts are found. The system displays a no-results message.
Exception flows	E1 (at step 4): The search keyword is empty. The system requests a keyword. E2 (at step 5): The search operation fails. The system displays an error message.
Business rules	BR-16: Searching forum posts requires login. BR-17: Hidden or removed posts do not appear in search results.
Non-functional constraints	Search results must be displayed clearly.

7.7 Use Case UC-07: Create Forum Post

Use case ID	UC-07
Use case name	Create Forum Post

Brief description	Allows a registered user to publish a post or story in the community forum.
Primary actor	Registered User
Secondary actors	Cloudinary
Priority	Must
Related requirements	FR-02.3
Trigger	The user selects Create Post.
Pre-conditions	The user is logged in.
Post-conditions	A new forum post is saved and published.
Main flow	1. The user opens the community forum. 2. The user selects Create Post. 3. The system displays the post creation form. 4. The user enters the post content. 5. The user optionally uploads an image. 6. The user submits the post. 7. The system validates the submitted information. 8. The system saves and publishes the post. 9. The system displays a success message and the new post.
Alternative flows	A1 (at step 5): The user does not upload an image. The system continues with text-only content.
Exception flows	E1 (at step 7): The post content is missing or invalid. The system displays validation messages. E2 (at step 8): The post cannot be saved. The system displays an error and does not publish it. E3 (at step 5): The image cannot be uploaded to Cloudinary. The system displays an error and keeps the entered content so that the user can try again.
Business rules	BR-16: Only logged-in users can create forum posts. BR-05: Published posts are subject to administrator moderation when they are reported.
Non-functional constraints	The post form must provide clear labels and validation feedback. Uploaded images must be handled securely.

7.8 Use Case UC-08: Comment and React to Posts

Use case ID	UC-08
Use case name	Comment and React to Posts
Brief description	Allows a registered user to comment on or react to an existing forum post.
Primary actor	Registered User
Secondary actors	None
Priority	Must
Related requirements	FR-02.5
Trigger	The user selects a post and chooses to comment or react.

Pre-conditions	The user is logged in and the selected post is available.
Post-conditions	The submitted comment or reaction is recorded and displayed.
Main flow	1. The user opens a forum post. 2. The system displays the post and available interactions. 3. The user chooses to comment or react. 4. If commenting, the user enters comment content. 5. The user submits the comment or reaction. 6. The system validates the submitted information. 7. The system records the interaction. 8. The system displays the updated post interaction.
Alternative flows	A1 (at step 3): The user reacts without writing a comment. The flow skips step 4 and the system records the reaction.
Exception flows	E1 (at step 6): The comment is empty or invalid. The system displays a validation message. E2 (at step 7): The system cannot save the interaction. It displays an error message.
Business rules	BR-16: Only logged-in users can comment on or react to posts.
Non-functional constraints	The interaction interface must provide clear feedback after submission. Interactions must be associated with the authenticated account.

7.9 Use Case UC-09: Share Forum Post

Use case ID	UC-09
Use case name	Share Forum Post
Brief description	Allows a registered user to obtain a shareable link to a forum post.
Primary actor	Registered User
Secondary actors	None
Priority	Must
Related requirements	FR-02.6
Trigger	The user selects Share on a forum post.
Pre-conditions	The user is logged in and the selected post is available.
Post-conditions	A shareable link to the selected post is provided.
Main flow	1. The user opens a forum post. 2. The system displays the post and its available actions. 3. The user selects Share. 4. The system generates or retrieves the post's shareable link. 5. The system provides the link to the user.
Alternative flows	A1 (at step 5): The user copies the link. The system makes the link available for copying.

Exception flows	E1 (at step 4): The system cannot generate or retrieve the link. It displays an error message.
Business rules	BR-17: A hidden or removed post cannot be opened through its share link. BR-16: Getting a share link requires login; opening the link to a public post does not.
Non-functional constraints	The share function must be easy to find and use. The generated link must direct users to the correct post.

7.10 Use Case UC-10: Create Fundraising Campaign

Use case ID	UC-10
Use case name	Create Fundraising Campaign
Brief description	Allows a campaign organizer to submit a new campaign for administrator review.
Primary actor	Campaign Organizer
Secondary actors	Cloudinary
Priority	Must
Related requirements	FR-03.1
Trigger	The organizer selects Create Campaign.
Pre-conditions	The organizer is logged in and has permission to create campaigns.
Post-conditions	A new campaign is saved with Pending Approval status, or with Draft status if the organizer saved a draft (A2).
Main flow	1. The organizer opens the campaign management page. 2. The organizer selects Create Campaign. 3. The system displays the campaign creation form. 4. The organizer enters the campaign title and story. 5. The organizer uploads a cover image. 6. The organizer enters the donation goal and deadline. 7. The organizer submits the campaign. 8. The system validates the required information. 9. The system saves the campaign with pending approval status. 10. The system displays a submission confirmation.
Alternative flows	A1 (at step 5): The organizer selects a different cover image before submission. The system uses the latest selected image. A2 (at step 7): The organizer selects Save Draft. The system saves the campaign with Draft status (BR-09); it is not public and can be edited and submitted later (UC-11 A4).
Exception flows	E1 (at step 8): Required fields are missing or invalid. The system displays validation messages and requests corrections. E2 (at step 9): The campaign cannot be saved. The system displays an error message.

	E3 (at step 5): The cover image cannot be uploaded to Cloudinary. The system displays an error, keeps the entered information and does not submit the campaign.
Business rules	BR-02: Campaigns must include a title, story, cover image, donation goal and deadline. BR-03: Campaigns must be approved before becoming publicly visible. BR-06: Campaign approval is an internal manual review, not official identity verification.
Non-functional constraints	The campaign form must provide clear instructions and validation feedback. Campaign information and uploaded images must be stored securely.

7.11 Use Case UC-11: Edit or Close Campaign

Use case ID	UC-11
Use case name	Edit or Close Campaign
Brief description	Allows an organizer to update or close an existing campaign they manage.
Primary actor	Campaign Organizer
Secondary actors	Cloudinary
Priority	Must
Related requirements	FR-03.2, FR-03.3
Trigger	The organizer selects a campaign and chooses Edit or Close.
Pre-conditions	The organizer is logged in and owns the selected campaign.
Post-conditions	The edited campaign is saved with a valid status under BR-09, or the campaign is Closed; no unauthorized change occurs.
Main flow	1. The organizer opens campaign management. 2. The system lists campaigns owned by the organizer. 3. The organizer selects a campaign. 4. The system shows its status and permitted actions.

	5. The organizer selects Edit for a Draft, Rejected, or Approved/Active campaign. 6. The system displays editable fields. 7. The organizer changes information and submits it. 8. The system validates ownership, inputs, and the current status. 9. The system saves Draft changes as Draft; for an edit to a field listed in BR-02 of an Approved/Active campaign, it sets the revised campaign to Pending Approval and removes it from public visibility until reapproved. 10. The system displays the resulting status.
Alternative flows	A1 (at step 5): The organizer selects Close for an Approved/Active campaign. After confirmation, the system sets Closed, blocks further donations, and retains donation history. A2 (at step 7): The organizer cancels editing; no changes are saved. A3 (at step 5): The organizer edits a Rejected campaign and saves it as Draft; a later submission moves it to Pending Approval. A4 (at step 5): The organizer selects Submit for a Draft campaign. The system validates the required fields (BR-02) and sets the campaign to Pending Approval (BR-09). A5 (Automatic closure): When an Approved/Active campaign reaches its deadline, the system automatically changes its status to Closed. The campaign remains publicly accessible in read-only mode, and its donation history is preserved.
Exception flows	E1 (at step 3): Campaign is not owned by the organizer; deny access and preserve data. E2 (at step 8): Missing/invalid fields; show validation errors and preserve existing data. E3 (at step 8): Campaign is Closed or the status changed concurrently; deny editing/closing and refresh status. E4 (at step 9): Save fails; show error and keep previous status and data.
Business rules	BR-02 (required fields); BR-03 (approval before publication); BR-09 (allowed status transitions); BR-10 (donations only to Approved/Active campaigns); BR-18 (organizers manage only their own campaigns).

Non-functional constraints	Only authorized organizers can edit or close their campaigns. The system must request confirmation before closing a campaign.

7.12 Use Case UC-12: Review Fundraising Campaign

Use case ID	UC-12
Use case name	Review Fundraising Campaign
Brief description	Allows an administrator to inspect a submitted campaign before making an approval decision.
Primary actor	Administrator
Secondary actors	None
Priority	Must
Related requirements	FR-04.1
Trigger	The administrator opens the campaign review queue.
Pre-conditions	The administrator is logged in.
Post-conditions	The selected campaign has been reviewed; its status remains Pending Approval until a decision is made in UC-13.
Main flow	1. The administrator opens the review queue. 2. The system lists Pending Approval campaigns. 3. The administrator selects a campaign. 4. The system displays its submitted details. 5. The administrator checks completeness and compliance with internal platform rules. 6. The administrator completes review and proceeds to UC-13 for a decision.
Alternative flows	A1 (at step 2): No campaigns are awaiting review. The system displays an empty review queue.
Exception flows	E1 (at step 3): The selected campaign is no longer available. The system informs the administrator and refreshes the queue.

	E2 (at step 4): Campaign information cannot be retrieved. The system displays an error.
Business rules	BR-03: Campaigns must be approved before becoming publicly visible. BR-04: Only administrators can approve or reject campaigns. BR-06: The review is internal and is not an official identity or background check.
Non-functional constraints	Campaign information must be accessible only to authorized administrators during review. The review interface must display campaign information clearly.

7.13 Use Case UC-13: Approve or Reject Campaign

Use case ID	UC-13
Use case name	Approve or Reject Campaign
Brief description	Allows an administrator to approve or reject a campaign after reviewing it.
Primary actor	Administrator
Secondary actors	None
Priority	Must
Related requirements	FR-04.2, FR-04.3
Trigger	The administrator selects an approval or rejection decision for a reviewed campaign.
Pre-conditions	The administrator is authenticated; the selected campaign is in Pending Approval and has been reviewed.
Post-conditions	The campaign changes from Pending Approval to Approved/Active or Rejected. An approved campaign becomes publicly visible; a rejected campaign remains

	unpublished, and its rejection reason is available to the organizer.
Main flow	1. The administrator opens a campaign awaiting a decision. 2. The system displays the campaign information. 3. The administrator selects Approve, or selects Reject and enters a reason (BR-15). 4. The system requests confirmation. 5. The administrator confirms the decision. 6. The system records the decision. 7. The system updates the campaign status. 8. If approved, the system makes the campaign publicly visible. 9. If rejected, the system keeps the campaign unpublished and shows the reason to the organizer. 10. The system displays a confirmation message.
Alternative flows	A1 (at step 4): The administrator cancels the decision. The system returns to campaign details without changing the status.
Exception flows	E1 (at step 1 or 5): Campaign is not in Pending Approval; reject the transition and refresh status. E2 (at step 6): Decision cannot be saved; preserve the previous status and display an error. E3 (at step 5): User lacks Administrator permissions; deny the action and preserve status. E4 (at step 5): Reject was selected without a reason. The system asks for the reason and does not change the status (BR-15).
Business rules	BR-03 (publication requires approval); BR-04 (administrator-only decisions); BR-06 (internal review only); BR-09 (allowed transitions); BR-15 (a rejection requires a reason).
Non-functional constraints	Only administrators can perform approval or rejection actions. The system must accurately record and display campaign status.

7.14 Use Case UC-14: Moderate Reported Forum Posts

Use case ID	UC-14

Use case name	Moderate Reported Forum Posts
Brief description	Allows an administrator to review reported posts and apply an appropriate moderation action.
Primary actor	Administrator
Secondary actors	None
Priority	Must
Related requirements	FR-04.4
Trigger	The administrator opens the reported posts section.
Pre-conditions	The administrator is logged in.
Post-conditions	The selected reported post is reviewed and the chosen moderation action is applied.
Main flow	1. The administrator opens the administration dashboard. 2. The administrator selects Reported Forum Posts. 3. The system displays reported posts. 4. The administrator selects a reported post. 5. The system displays the post content and report information. 6. The administrator reviews the post. 7. The administrator selects Hide post or Remove post. 8. The system applies the selected action. 9. The system updates the post's moderation status. 10. The system displays a confirmation message.
Alternative flows	A1 (at step 3): No reported posts are available. The system displays an empty-state message. A2 (at step 7): The administrator selects No action required. The system leaves the post unchanged.
Exception flows	E1 (at step 4): The reported post is no longer available. The system informs the administrator and refreshes the list. E2 (at step 8): The moderation action cannot be applied. The system displays an error.
Business rules	BR-05: Administrators are responsible for moderating reported forum posts according to platform rules. BR-17: A hidden or removed post is no longer shown publicly. BR-01: Only administrators can perform moderation actions.
Non-functional constraints	Moderation functions must be restricted to authorized administrators. The system must clearly display the result of a moderation action.

7.15 Use Case UC-15: Manage User Accounts

Use case ID	UC-15
Use case name	Manage User Accounts
Brief description	Allows an administrator to view and manage user accounts and assigned roles.

Primary actor	Administrator
Secondary actors	None
Priority	Must
Related requirements	FR-01.5
Trigger	The administrator opens the user management section.
Pre-conditions	The administrator is logged in and has permission to manage user accounts.
Post-conditions	Selected account information or role assignment is updated if a change is made.
Main flow	1. The administrator opens the administration dashboard. 2. The administrator selects User Management. 3. The system displays available user accounts. 4. The administrator selects a user account. 5. The system displays the selected account information. 6. The administrator changes the roles assigned to the account. 7. The administrator confirms the action. 8. The system validates the administrator's permission. 9. The system saves the changes. 10. The system displays a confirmation message.
Alternative flows	A1 (at step 6): The administrator only views account information and makes no changes. The system returns to the account list.
Exception flows	E1 (at step 4): The selected account cannot be found. The system displays an error message. E2 (at step 8): The administrator lacks permission. The system denies the request. E3 (at step 9): The changes cannot be saved. The system displays an error.
Business rules	BR-01: Users access system features according to their assigned roles; only administrators can change role assignments.
Non-functional constraints	User account information must be protected from unauthorized access. Administrative actions must be clearly confirmed by the system.

7.16 Use Case UC-16: Make Simulated Donation

Use case ID	UC-16
Use case name	Make Simulated Donation
Brief description	Allows a donor to make a simulated donation to an approved campaign using the demo wallet.
Primary actor	Donor
Secondary actors	None

Priority	Must
Related requirements	FR-05.1, FR-05.2
Trigger	The donor selects Donate on an approved campaign.
Pre-conditions	The donor is logged in and the selected campaign is Approved/Active (BR-10).
Post-conditions	After a successful donation, the simulated donation is recorded with its optional message and anonymity choice, the demo wallet balance is reduced, and the campaign total and progress are updated.
Main flow	1. The donor opens an Approved/Active campaign. 2. The system displays campaign details and donation progress. 3. The donor selects Donate. 4. The system displays step 1, Amount (SCR-01_UC-16_Donation_Form), with the demo wallet balance. 5. The donor selects or enters an amount, may add a message, may choose to donate anonymously (BR-13), and continues. 6. The system checks the amount (BR-12) and displays step 2, Demo wallet (SCR-02_UC-16_Demo_Wallet), with the available balance and the balance after the donation. 7. The donor continues to the review step. 8. The system displays step 3, Review and confirm (SCR-03_UC-16_Review_Confirm), with the campaign, amount, display name and message. 9. The donor accepts the terms (a completed donation cannot be cancelled, BR-14) and confirms the donation. 10. The system validates the amount, the demo wallet balance and the campaign status again. 11. The system records the simulated donation. 12. The system deducts the amount from the donor's demo wallet. 13. The system updates the campaign amount raised and progress. 14. The system displays the success screen (SCR-04_UC-16_Donation_Successful) with the donation details and the updated progress.
Alternative flows	A1 (at steps 6–9): The donor goes back and changes the amount. The system uses the latest amount entered and recalculates the balance after the donation. A2 (at steps 5–9): The donor selects Back to campaign. The system returns to the campaign page without recording a donation.

Exception flows	E1 (at step 6): The amount is below 10,000 VND or is not a valid number (BR-12). The system displays a validation message and stays on step 1. E2 (at step 6): The amount exceeds the demo wallet balance (BR-12). The system displays SCR-02_UC-16_Demo_Wallet_Insufficient_Balance; the donor cannot continue until a smaller amount is entered. E3 (at step 11): The donation cannot be recorded. The system displays SCR-04_UC-16_Donation_Failed, records no donation and does not deduct the wallet balance (BR-11). E4 (at step 13): The progress cannot be updated. The system cancels the donation record and the wallet deduction and displays SCR-04_UC-16_Donation_Failed (BR-11). E5 (at step 10): The campaign is no longer Approved/Active or its deadline has passed (BR-10). The system rejects the donation, displays an appropriate error message, and makes no changes to the donor's demo wallet, donation records, or campaign progress.
Business rules	BR-07: Donations are simulated through the demo wallet; no real-money transaction is processed. BR-08: Campaign totals and progress bars are updated after each successful simulated donation. BR-10: Only Approved/Active campaigns can receive donations. BR-11: A failed donation changes nothing. BR-12: Minimum amount and wallet balance limit. BR-13: Optional message and anonymous donation. BR-14: A completed donation cannot be cancelled.
Non-functional constraints	The donation operation is available only to the authenticated donor (NFR-02). A test user completes the donation flow within 3 minutes (NFR-03). The system must clearly identify the donation as simulated.

7.17 Use Case UC-17: View Donation Progress

Use case ID	UC-17
Use case name	View Donation Progress
Brief description	Allows a guest or logged-in user to view the amount raised and progress of an approved fundraising campaign.
Primary actor	Guest
Secondary actors	None
Priority	Must
Related requirements	FR-05.2
Trigger	The guest opens the detail page of an approved campaign.

Pre-conditions	The selected campaign has been approved and is publicly visible.
Post-conditions	The latest available donation total and progress are displayed; no information is modified.
Main flow	1. The guest opens the campaign page. 2. The system retrieves the campaign's donation information. 3. The system calculates or retrieves donation progress. 4. The system displays the amount raised, donation goal and progress bar. 5. The guest views the fundraising progress.
Alternative flows	A1 (at step 2): The campaign has no donations. The system displays an amount raised of zero and the corresponding progress. A2 (at step 4): The guest refreshes the page. The system retrieves and displays the latest available donation information.
Exception flows	E1 (at step 2): Donation information cannot be retrieved. The system displays an error message. E2 (at step 3): Progress cannot be calculated. The system displays available campaign information and an appropriate message.
Business rules	BR-08: Campaign donation totals and progress bars must be updated after each successful simulated donation. Donation progress is the amount raised shown as a percentage of the donation goal (Glossary).
Non-functional constraints	Campaign progress information must meet applicable performance requirements in NFR-01. The progress bar must meet NFR-07 (accessibility). Progress is updated when the page is loaded or refreshed; real-time push updates are out of scope.

7.18 Use Case UC-18: View Donation History

Use case ID	UC-18
Use case name	View Donation History
Brief description	Allows a donor to view the simulated donations they have made from their demo wallet.
Primary actor	Donor
Secondary actors	None
Priority	Must
Related requirements	FR-05.3
Trigger	The donor selects View my donations after a donation or My donations in the account menu.
Pre-conditions	The donor is logged in.

Post-conditions	The donor's recorded donations are displayed; no information is modified.
Main flow	1. The donor opens the My donations page. 2. The system retrieves the simulated donations recorded for the donor. 3. The system displays the total amount donated, the number of campaigns supported and the number of donations. 4. The system lists each donation with its date and time, campaign, amount, method and status. 5. The donor reviews the donation history.
Alternative flows	A1 (at step 3): The donor has no recorded donations. The system displays an empty-state message. A2 (at step 5): The donor selects Explore campaigns. The system displays the campaign list.
Exception flows	E1 (at step 2): The donation history cannot be retrieved. The system displays an error message.
Business rules	BR-07: Donations must be simulated through the demo wallet; no real-money transaction is processed. BR-11: Failed attempts create no donation record, so only successful donations are listed. BR-18: A donor sees only their own donation history.
Non-functional constraints	Donation history must be accessible only to the authenticated donor. Amounts must be displayed in the same format as on the campaign pages.

8 Prototype / Mockups

Link to the overall Figma project:
https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF/All-Screen?node-id=0-1&t=u3XWexHBX75sZHYX-1
Link to the interactive prototype:
https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF/All-Screen?node-id=0-1&p=f&t=u3XWexHBX75sZHYX-0

Screens	Related use case(s)	Mockup / link
SCR-01_UC-01_Register	UC-01	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=19-1036
SCR-01_UC-02_Login	UC-02	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=19-1037
SCR-01_UC-03_Profile_View SCR-02_UC-03_Profile_Edit	UC-03	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1332
SCR-01_UC-04_Campaign_List SCR-02_UC-04_Campaign_Detail	UC-04	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1333
SCR-01_UC-05_Forum_Post_List SCR-02_UC-05_Forum_Post_Detail SCR-03_UC-05_Empty_Forum (A1) SCR-04_UC-05_Forum_Load_Error (E1)	UC-05	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1334

SCR-05_UC-05_Forum_Loading		
SCR-01_UC-06_Forum_Search SCR-02_UC-06_Search_Results SCR-03_UC-06_No_Search_Results (A1) SCR-04_UC-06_Search_Keyword_Error (E1) SCR-05_UC-06_Search_Failed (E2) SCR-06_UC-06_Search_Loading	UC-06	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1335
SCR-01_UC-07_Create_Post_Form SCR-02_UC-07_Post_With_Image SCR-03_UC-07_Post_Published SCR-04_UC-07_Post_Validation_Error (E1) SCR-05_UC-07_Post_Publish_Failed (E2) SCR-06_UC-07_Sign_In_Required	UC-07	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1336
SCR-01_UC-08_Post_Interactions SCR-02_UC-08_Comment_Posted SCR-03_UC-08_Reaction_Added (A1) SCR-04_UC-08_Comment_Validation_Error (E1) SCR-05_UC-08_Interaction_Save_Failed (E2)	UC-08	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1337

SCR-06_UC-08_Sign_In_Required		
SCR-01_UC-09_Post_Share_Action SCR-02_UC-09_Share_Link SCR-03_UC-09_Link_Copied (A1) SCR-04_UC-09_Share_Link_Error (E1)	UC-09	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1338
SCR-01_UC-10_Create_Campaign SCR-02_UC-10_Submit_Campaign_for_Approval	UC-10	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1339
SCR-01_UC-11_My_Campaigns SCR-02_UC-11_Edit_Campaign SCR-03_UC-11_Campaign_Management_Detail (A1 Close)	UC-11	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1340
SCR-01_UC-12_Campaign_Approval_Dashboard SCR-02_UC-12_Review_Campaign_Detail	UC-12	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1341
SCR-01_UC-13_Approval_Decision	UC-13	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1342
SCR-01_UC-14_Reported_Posts SCR-02_UC-14_Reported_Post_Review SCR-03_UC-14_Post_Hidden SCR-04_UC-14_Post_Removed SCR-05_UC-14_No_Action_Required (A2) SCR-06_UC-14_Empty_Reported_Posts (A1) SCR-07_UC-14_Reported_Post_Unavailable (E1)	UC-14	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1343

SCR-08_UC-14_Moderation_Failed (E2) SCR-09_UC-14_Permission_Denied		
SCR-01_UC-15_User_Management SCR-02_UC-15_Account_Detail SCR-02_UC-15_Account_Detail_Confirm_Role_Change SCR-02_UC-15_Account_Detail_Roles_Updated	UC-15	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1344
SCR-01_UC-16_Donation_Form SCR-02_UC-16_Demo_Wallet SCR-02_UC-16_Demo_Wallet_Insufficient_Balance (E2) SCR-03_UC-16_Review_Confirm SCR-04_UC-16_Donation_Successful SCR-04_UC-16_Donation_Failed (E3, E4)	UC-16	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1345
SCR-01_UC-17_Campaign_Progress	UC-17	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1346
SCR-01_UC-18_My_Donations	UC-18	https://www.figma.com/design/ekg8MUUWDCOINuOA4Bt7KF?node-id=39-1347

Notes on the screens. (1) The table lists the 62 desktop frames (1440 px). Every desktop frame also has a 360 px mobile version on the same Figma page, whose name ends in _Mobile (62 mobile frames, 124 frames in total). (2) SCR-01_UC-04 and SCR-02_UC-04 are drawn in the logged-in state; a guest, the primary actor of UC-04, sees the same content with Log in / Sign up in the navigation bar. (3) Accessibility check: an automated contrast audit of the account and donation screens (UC-01 to UC-04 and UC-15 to UC-18) found no text below the 4.5:1 ratio after the fixes of 29/10 and 08/11; form fields use the library Input component with a visible label, and the

library Button has a focus state. The screens of UC-05 to UC-14 have not been audited yet; they are checked against NFR-07 during the UI design of PA3 (WBS 3.5.4).

9 Requirements Traceability Matrix

This matrix is extended in PA3 (design elements) and PA4 (test cases).

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-01.1	UC-01	SCR-01_UC-01_ Register	To be defined in PA3	To be defined in PA4
FR-01.2	UC-02	SCR-01_UC-02_ Login	To be defined in PA3	To be defined in PA4
FR-01.3	UC-03	SCR-01_UC-03_ Profile_View, SCR-02_UC-03 _Profile_Edit	To be defined in PA3	To be defined in PA4
FR-01.4	UC-02 (step 7); login pre-conditions of UC-03, UC-06 to UC-16 and UC-18; UC-13 E3, UC-15 E2	SCR-06_UC-07_ Sign_In_Required, SCR-06_UC-08 _Sign_In_Required, SCR-09_UC-14_ Permission_Denied	To be defined in PA3	To be defined in PA4
FR-01.5	UC-15	SCR-01_UC-15_ User_Management, SCR-02_UC-15_ Account_Detail (with Confirm_Role_Change and Roles_Updated)	To be defined in PA3	To be defined in PA4
FR-02.1	UC-05	SCR-01 to SCR-05 of UC-05	To be defined in PA3	To be defined in PA4
FR-02.2	UC-06	SCR-01 to SCR-06 of UC-06	To be defined in PA3	To be defined in PA4

FR-02.3	UC-07	SCR-01 to SCR-06 of UC-07	To be defined in PA3	To be defined in PA4
FR-02.4	No use case yet (review item 10, deferred to PA3)	–	To be defined in PA3	To be defined in PA4
FR-02.5	UC-08	SCR-01 to SCR-06 of UC-08	To be defined in PA3	To be defined in PA4
FR-02.6	UC-09	SCR-01 to SCR-04 of UC-09	To be defined in PA3	To be defined in PA4
FR-03.1	UC-10	SCR-01_UC-10_Create_Campaign, SCR-02_UC-10_Submit_Campaign_for_Approval	To be defined in PA3	To be defined in PA4
FR-03.2	UC-11	SCR-01_UC-11_My_Campaigns, SCR-02_UC-11_Edit_Campaign	To be defined in PA3	To be defined in PA4
FR-03.3	UC-11 (A1)	SCR-03_UC-11_Campaign_Management_Detail	To be defined in PA3	To be defined in PA4
FR-03.4	UC-04	SCR-01_UC-04_Campaign_List, SCR-02_UC-04_Campaign_Detail	To be defined in PA3	To be defined in PA4
FR-04.1	UC-12	SCR-01_UC-12_Campaign_Approval_Dashboard, SCR-02_UC-12_Review_Campaign_Detail	To be defined in PA3	To be defined in PA4
FR-04.2	UC-13	SCR-01_UC-13_Approval_Decision	To be defined in PA3	To be defined in PA4
FR-04.3	UC-04, UC-13	SCR-01_UC-04_Campaign_List,	To be defined in PA3	To be defined in PA4

		SCR-01_UC-11_My_Campaigns		
FR-04.4	UC-14	SCR-01 to SCR-09 of UC-14	To be defined in PA3	To be defined in PA4
FR-05.1	UC-16	SCR-01 to SCR-04 of UC-16	To be defined in PA3	To be defined in PA4
FR-05.2	UC-17, UC-04, UC-16 (step 13)	SCR-01_UC-17_Campaign_Progress, SCR-02_UC-04_Campaign_Detail, SCR-04_UC-16_Donation_Successful	To be defined in PA3	To be defined in PA4
FR-05.3	UC-18	SCR-01_UC-18_My_Donations	To be defined in PA3	To be defined in PA4
FR-06, FR-07, FR-08	No use case in this baseline (Should-have)	–	To be defined in PA3	To be defined in PA4
FR-09	No use case in this baseline (Could-have, not baselined)	–	To be defined in PA3	To be defined in PA4

10 Requirements Review

Review Summary: The team reviewed requirement completeness, consistency, testability, prototype coverage and traceability. Findings and resolutions are documented in the table below.

Participants: Trần Anh Khoa (project manager / tester), Nguyễn Trần Phương Thúy (business analyst), Hồng Gia Bảo (designer), Lê Nho Duy Phước (developer). The review activities and the reviewers of each activity are listed in the first table below.

Method: walkthroughs in the weekly status meetings, and document inspections against a checklist: (1) consistency with the scope, priorities and constraints of PA0 and PA1; (2) completeness: every Must-have feature has functional requirements, business rules, acceptance criteria, use cases and screens; (3) every non-functional requirement states a metric, its conditions and how it is verified; (4) the use case model uses actors, associations, include, extend and generalisation correctly; (5) every use case has pre-conditions, post-conditions and numbered main, alternative and exception flows; (6) the prototype shows nothing that the specification does not state, and the reverse; (7) requirements, use cases and screens trace to each other in the RTM.

Result: 38 issues recorded: 10 were fixed in earlier drafts, 25 are fixed in this baseline version, and 3 are deferred to PA3 as open questions (see the baseline statement after the issue table).

Date	Reviewer(s)	Reviewed Content
23/10/2026	Khoa, Thúy	Initial requirements and RTM review (W04).
29/10/2026	Phước	Donation flow (FL-05) vs. PA1 scope.
01/11/2026	All four members	Requirements draft and initial RTM review (W05).
03/11/2026	Phước	SRS consistency with PA0, PA1, and project constraints.
08/11/2026	All four members	Figma prototype, UC naming, flows, and diagram consistency (W06).
08/11/2026	Phước	SRS, prototype, and use case cross-check.
10/11/2026	Bảo	NFR and RTM review.
10/11/2026	Thúy	Prototype coverage and UI accessibility review.
11/11/2026	Khoa	Section 6, UC-10–UC-13, and FR/AC/RTM mappings.

#	Issue found	Requirement(s)	Resolution
A. Consistency with PA0 and PA1			
1	The first donation mockups (flow FL-05) used an external payment gateway (VietQR, e-wallet, card) as the payment step, although PA1 makes the demo wallet the Must-have method (F05) and the sandbox gateway only a Could-have (F09). They also showed real-time wording, email/PDF receipts and "fund disbursement". (Found 29/10.)	F05, F09 C-02, C-04 BR-07	**Fixed (29/10).** The payment step became "Demo wallet" with the balance before and after the donation, an Insufficient Balance screen was added, and receipts, real-time wording and "disbursement" were removed (expense reports belong to F06).
2	Sections 2–3 called F04 "case/campaign verification" and described campaigns as "verified". PA1 Section 2.1 says that approval does not determine whether a campaign is legitimate, and that identity and legal verification are out of scope. The Administrator actor in Section 6.2 was still described as a user who "verifies fundraising cases and organizers". (Found 03/11.)	Scope table (F04) Glossary P2, G2 C-03, A-03 Section 6.2 BR-06	**Fixed (03/11 and 10/11).** Sections 2–3 now say "campaign approval" and "reviewed and approved"; C-03 forbids presenting approval as verification; the prototype badge reads "Approved campaign". The Administrator description was reworded in the revision of 10/11.
3	Sections 2–3 quoted figures and sources that differ from the submitted PA1: 360 person-hours over 10 weeks (PA1: 367 over 11 weeks), an out-of-scope list taken from PA0 instead of PA1 Section 2.4, and progress "updated when a page is loaded" (PA1 F05: right after each donation and on page reload). (Found 03/11.)	C-04, C-06 Out-of-scope list Glossary (Donation progress)	**Fixed (03/11).** Values and wording were aligned with PA1, and each constraint now cites its PA1 section.

4	FR-01 listed the roles "Admin, Campaign Manager, Member" and FR-03 spoke of "Campaign Managers". PA1 F01 and BR-01 define the roles Donor, Organizer, Volunteer and Admin. (Found 03/11; wording quoted from the draft of 08/11.)	FR-01, FR-03 BR-01	**Fixed (10/11).** FR-01 and FR-03 were rewritten with the four PA1 roles.
5	The feature references in the FR table did not match the PA1 feature list (found 03/11). The draft of 08/11 still had no requirement for F04 (campaign approval and moderation, Must-have), marked the marketplace requirement (F10) as Should although PA1 gives it Could, and used mixed labels in the Feature and Priority columns ("F01", "F03 – Fundraising", "Should Have").	FR-04 FR for F10 F04, F10	**Fixed (10/11).** FR-04 was added for F04, the F10 requirement was set to Could, feature names follow PA1 Section 2.4, and the MoSCoW labels are uniform.
B. Functional requirements			
6	After the revision of 10/11, FR-05 only covers viewing the donation history. No functional requirement says that a donor can make a simulated donation or that progress is updated, which is the core of F05. UC-16 and UC-17 still carry the note "FR-05 (as listed in the Use Case List; verify mapping)", and UC-18 points to FR-06, the volunteer task board. (Found 11/11.)	FR-05, FR-06 F05 UC-16, UC-17, UC-18 AC-F05-01 to AC-F05-06	**Fixed in this baseline.** FR-05 now groups the three F05 requirements: FR-05.1 simulated donation from the demo wallet, FR-05.2 amount raised and progress bar updated right after each successful donation and on page reload, FR-05.3 donation history. UC-16 maps to FR-05.1, UC-17 to FR-05.2 and UC-18 to FR-05.3; the notes were removed.

7	Each Must-have feature had exactly one FR that combined several capabilities (FR-01: register, log in, edit profile and role-based access). A single capability could not be traced or tested on its own, and the requirements were not really grouped by feature. (Found 11/11.)	FR-01 to FR-05	**Fixed in this baseline.** Each Must-have FR stays as a group and is split into atomic sub-requirements: FR-01.1–FR-01.5, FR-02.1–FR-02.6, FR-03.1–FR-03.4, FR-04.1–FR-04.4 and FR-05.1–FR-05.3. Existing clauses were separated; the only additions come from items 6, 9, 20 and 22. The Use Case List and the RTM refer to the sub-requirement IDs.
8	Section 2.1 states that the Should-have features F06, F07 and F08 are specified at FR level, but only F08 had a functional requirement. F06 (expense reporting) and F07 (email notifications) had none. (Found 11/11.)	Section 2.1 F06, F07, F08	**Fixed in this baseline.** FR-06 (F06) and FR-07 (F07) were added with the wording of PA1 Section 2.4. The volunteer task board is now FR-08 and the marketplace FR-09 (Could; listed only, not baselined). In FR-07, PA1's "verification results" is written as "campaign approval results" to stay consistent with C-03.
9	FR-04 gave administrators "authorized user management actions", while the Use Case List and the RTM trace user management (UC-15) to FR-01. UC-15 itself only says "an available account management action". The requirement was in the wrong group and did not name the action. (Found 11/11.)	FR-01, FR-04 UC-15	**Fixed in this baseline.** User management moved to FR-01.5 and is stated concretely: an administrator can view user accounts and change the roles assigned to a user, as drawn in the UC-15 screens. UC-15 step 6 names the action.

10	FR-02 requires that users can edit and delete their own posts. This comes from PA1 WBS 5.5.1, not from PA1 Section 2.4 F02 as the Source column says, and no use case, acceptance criterion or screen covers it. (Found 11/11.)	FR-02 (FR-02.4)	**Deferred to PA3.** Kept as FR-02.4 with the source corrected to PA1 WBS 5.5.1. The RTM shows "no use case yet". The use case, acceptance criteria and screen will be added through the PA3 change list.
11	UC-14 and FR-04 handle "reported" forum posts, but no requirement, use case or screen lets a user report a post, so a report can never be created by the system as specified. (Found 11/11.)	FR-04 (FR-04.4) BR-05 UC-14 AC-F04-03	**Deferred to PA3.** The team has not yet decided who may report a post and what a report contains. The question is carried to the PA3 change list; FR-04.4 is baselined for the moderation side only.
12	The SRS does not say which role a newly registered account receives, or how a user obtains the Organizer role. UC-01 creates "a user account", and UC-10 only requires that the organizer "has permission to create campaigns". (Found 11/11.)	FR-01 (FR-01.1, FR-01.4) BR-01 UC-01, UC-10, UC-15	**Deferred to PA3.** No decision has been made. The question is carried to the PA3 change list. In this baseline the only specified way to change roles is the administrator action in FR-01.5.
C. Non-functional requirements			
13	NFR-01 required a response within 1 s at P95 under 200 concurrent users for "real-time pages". This conflicts with C-04 (no real-time push) and cannot be reached on free-tier hosting (OE-04, C-05). (Found 03/11; wording quoted from the draft of 08/11.)	NFR-01 C-04, C-05 OE-04	**Fixed (10/11).** NFR-01 now requires at least 95% of campaign list and campaign detail requests within 3 s under 20 concurrent users, verified with JMeter.

14	The scalability target (100,000 active users below 70% CPU) and the availability target (99.5% per month with automatic daily backups) could not be verified in the free-tier environment. (Found 03/11; wording quoted from the draft of 08/11.)	NFR-04, NFR-05 C-05	**Fixed (10/11).** NFR-04 now requires 99% of requests without server error under 50 concurrent users, and NFR-05 requires 99% availability over a 24-hour monitoring period. The same revision completed NFR-03, which ended in "… SUS score of .", and removed a duplicated ID (two requirements were numbered NFR-05).
15	UC-02 still cites "HTTPS/TLS 1.3, as specified in NFR-02", but the revised NFR-02 only requires HTTPS. UC-06 and UC-16 refer to "applicable performance requirements" and NFR-01, although NFR-01 only measures campaign list and campaign detail requests. (Found 11/11.)	NFR-01, NFR-02 UC-02, UC-06, UC-16	**Fixed in this baseline.** UC-02 cites NFR-02 without "TLS 1.3". UC-06 no longer cites a performance requirement. UC-16 cites NFR-02 (protected endpoints) and NFR-03 (donation flow within 3 minutes).
16	NFR-01 and NFR-04 do not say whether the first request after the free-tier back end has been idle is counted. OE-04 states that this request can be slow, so the test result is ambiguous. (Found 11/11.)	NFR-01, NFR-04 OE-04	**Fixed in this baseline.** Both requirements now state the condition "measured after the back end has been woken up by a warm-up request".
17	No NFR covers responsive layout or accessibility. OE-01 and OE-02 commit to four browsers and a minimum width of 360 px, UC-05 and UC-17 ask for a "responsive" and "accessible" interface, and the PA2 template asks for a contrast, label and keyboard check. (Found 11/11.)	OE-01, OE-02 UC-05, UC-17 NFR-06, NFR-07 (new)	**Fixed in this baseline.** NFR-06 Compatibility (no horizontal scrolling at 360 px and 1440 px in the two latest versions of Chrome, Edge, Firefox and Safari) and NFR-07 Accessibility (text contrast of at least 4.5:1, a visible label on every form field, keyboard operation with a visible focus) were

			added. UC-05 and UC-17 cite them.
18	C-10 says that the handling of personal data is "detailed in Sections 5.2–5.3", but these sections only contain the password hashing rule in NFR-02. (Found 11/11.)	C-10 NFR-02	**Fixed in this baseline.** The source of C-10 now reads "Team decision; password storage is specified in NFR-02".
D. Business rules and acceptance criteria			
19	The use cases depended on campaign states and status rules that the business rule table did not define (found 03/11). The draft of 08/11 then named the states in BR-09, but not the allowed transitions or who may trigger them.	BR-09 UC-10, UC-11, UC-13	**Fixed (10/11).** BR-09 lists every transition (Draft → Pending Approval → Approved/Active or Rejected; Rejected → Draft; Approved/Active → Closed), the visibility of each state and the rejection of invalid transitions.
20	BR-09 starts every campaign in Draft and the prototype has a Save Draft button, but UC-10 saved a new campaign directly as Pending Approval and no flow submitted a draft. BR-09 also said "substantive fields" without naming them, and AC-F03-02 ignored the re-approval rule. (Found 11/11.)	BR-09 FR-03 (FR-03.1, FR-03.2) UC-10, UC-11 AC-F03-02	**Fixed in this baseline.** UC-10 A2 (Save Draft) and UC-11 A4 (submit a draft for approval) were added. BR-09 says "any field listed in BR-02" instead of "substantive fields". AC-F03-02 is limited to Draft campaigns; AC-F03-08 covers editing an Approved/Active campaign and AC-F03-09 covers Save Draft.

21	The donation screens apply rules that the SRS did not contain: a minimum of 10,000 VND per donation, an optional message of up to 200 characters, anonymous donation, and "simulated donations cannot be cancelled". AC-F05-03 rejected zero, negative and above-balance amounts, but not amounts below the minimum. (Found 08/11.)	FR-05 (FR-05.1) UC-16 AC-F05-03 BR-12, BR-13, BR-14 (new)	**Fixed in this baseline.** BR-12 (at least 10,000 VND and not above the wallet balance), BR-13 (optional message of up to 200 characters; anonymous donations shown as "Anonymous") and BR-14 (a completed simulated donation cannot be cancelled; refunds are out of scope in PA1 Section 2.4) were added. AC-F05-03 uses the minimum amount, and AC-F05-09 covers anonymous donations.
22	The approval screen requires a reason when a campaign is rejected ("Reason for rejection *"), and UC-11 lets the organizer revise a rejected campaign, but FR-04, UC-13 and AC-F04-02 did not mention a reason. (Found 11/11.)	FR-04 (FR-04.2) UC-13 AC-F04-02 BR-15 (new)	**Fixed in this baseline.** BR-15 was added: rejecting a campaign requires a reason, which is shown to the organizer. UC-13 steps 3 and 9 and the new exception E4, AC-F04-02 and the new AC-F04-07 were updated accordingly.
23	Several use cases state rules in free text that are not in the business rule table or are not cited by ID: guests can only browse (UC-05), hidden or removed posts are not shown (UC-05, UC-06, UC-09), ownership of campaigns and of the donation history (UC-11, UC-18), "only approved campaigns can receive donations" (which is BR-10), and the atomicity rule BR-11 that	BR-10, BR-11 UC-05 to UC-09, UC-11, UC-14 to UC-16, UC-18 BR-16, BR-17, BR-18 (new)	**Fixed in this baseline.** BR-16 (guest access), BR-17 (moderated posts are not public) and BR-18 (users act only on their own profile, posts, campaigns and donation history) were added. BR-11 was extended with "a failed donation creates no donation record and no wallet deduction". Every rule in the use case tables is now cited by ID.

	UC-16 applies in E2 and E3. (Found 11/11.)		
24	Three campaign situations are not specified: what happens when the deadline has passed, whether donations are still accepted after the goal has been reached, and whether a Closed campaign stays visible to the public (BR-09 says both "only Approved/Active campaigns are publicly visible" and "Closed campaigns remain read-only"). (Found 11/11.)	BR-09, BR-10 FR-03, FR-05 UC-11, UC-16	Fixed in this baseline. BR-09 and BR-10 were clarified to define campaign expiration, donation eligibility after reaching the goal, and Closed campaign visibility. UC-11 and UC-16 were updated to reflect these rules consistently.
25	Acceptance criteria were missing or vague for Must-have behaviour: none for the donation history (UC-18), none for a successful role change (UC-15), AC-F04-03 said "a moderation action" without naming it, and AC-F02-03 let any user search although UC-06 requires login. (Found 11/11.)	AC-F01, AC-F02-03, AC-F04-03, AC-F05 UC-06, UC-14, UC-15, UC-18	**Fixed in this baseline.** AC-F01-08 (role change), AC-F05-07 and AC-F05-08 (donation history; only the donor's own successful donations) were added. AC-F04-03 names the actions hide and remove. AC-F02-03 starts with "a logged-in user".
E. Use case model			
26	The use case diagram in the draft of 08/11 had wrong associations: Donor had no use case; Make Simulated Donation and View Donation Progress were attached to Campaign Organizer; Create Campaign and Edit/Close Campaign were attached to Administrator; Register Account, Moderate Reported Forum Posts and View Donation History were missing; several names differed from the Use Case List. (Found 08/11.)	Section 6.1 UC-01, UC-10, UC-11, UC-14, UC-16, UC-17, UC-18	**Fixed (10/11).** The diagram was redrawn with all 18 use cases, their IDs and the primary actors of the Use Case List.

27	The redrawn diagram still has modelling errors. (a) Log In «include» Register Account: logging in does not always include registering, and Register Account has no actor although its primary actor is Guest. (b) Donor has no generalisation to Registered User although Section 6.2 says so, while the Administrator arrow is not mentioned in the text. (c) Registered User is not connected to browsing (UC-04, UC-05, UC-17), and UC-06, whose primary actor is Registered User, can only be reached as an extension of the guest's UC-05. (d) There is no system boundary, and Cloudinary, the secondary actor of UC-07, UC-10 and UC-11, is not drawn. (Found 11/11; point (d) first noted 08/11.)	Sections 6.1, 6.2 UC-01, UC-02, UC-05, UC-06, UC-07, UC-10, UC-11	**Fixed in this baseline.** (a) The «include» was removed and Guest is associated with UC-01. (b) Donor generalises Registered User, and Section 6.2 lists Donor, Campaign Organizer and Administrator as specialised roles. (c) Registered User generalises Guest and is associated with UC-06; the «extend» to UC-05 keeps the condition "user is logged in". (d) The boundary "ImpactHub" and the actor Cloudinary were added.
28	Fourteen use cases list "ImpactHub System" as secondary actor. The system being specified is not an actor (UC-18 already uses "None"). Cloudinary is used as a secondary actor but is missing from the Actors table. (Found 11/11.)	Section 6.2 UC-01 to UC-06, UC-08, UC-09, UC-12 to UC-17	**Fixed in this baseline.** "ImpactHub System" was replaced by "None", and Cloudinary (external image storage, OE-05) was added to the Actors table.
F. Use case specifications			
29	The "My donations" screen of the prototype had no use case and no requirement. (Found 08/11.)	UC-18 FR-05 (FR-05.3)	**Fixed (10/11).** UC-18 View Donation History and a matching requirement were added; the Donor actor and the Donors stakeholder mention the history. The FR

			reference of UC-18 is corrected under item 6.
30	In the draft of 10/11 the content of UC-11, UC-12 and UC-13 sits in the wrong fields. In UC-11 the main flow is in "Post-conditions", the alternative flows in "Main flow", the exception flows in "Alternative flows" and the rule list in "Exception flows". UC-12 has a second main flow in "Post-conditions". In UC-13 the trigger holds the pre-condition, "Alternative flows" holds exceptions and "Exception flows" holds the rule list. The earlier pre-conditions and post-conditions were overwritten. (Found 11/11.)	UC-11, UC-12, UC-13	**Fixed in this baseline.** The three tables were rebuilt so that each field holds its own content. The older duplicate main flow of UC-12 was removed, and the trigger, pre-conditions, post-conditions and alternative flow A1 of UC-13 were restored from the draft of 08/11.
31	UC-16 describes one form and one confirmation, while the prototype has three steps (Amount, Demo wallet, Review and confirm) with separate success, failure and insufficient-balance screens. E1 mixed an invalid amount with an insufficient balance, E3 was vague ("must not report the operation as successful unless ..."), and no exception covered a campaign that is no longer Approved/Active although AC-F05-04 tests it. (Found 08/11.)	UC-16 BR-10, BR-11, BR-12 AC-F05-04	**Fixed in this baseline.** The main flow was rewritten in 14 steps that name the screens. Exceptions: E1 amount below the minimum, E2 insufficient balance, E3 donation cannot be recorded, E4 progress cannot be updated (the whole donation is cancelled, BR-11), E5 campaign no longer Approved/Active.

32	UC-12 step 6 lets the administrator review "supporting information voluntarily submitted", but UC-10 and BR-02 give the organizer no field in which to submit such information. (Found 11/11.)	UC-10, UC-12 BR-02	Fixed in this baseline. The unsupported reference to voluntarily submitted supporting information was removed from UC-12. The administrator now reviews only the campaign information specified in UC-10 and BR-02. No additional upload fields or requirements were introduced.
33	Smaller defects in the flows: UC-01 A1 (email already registered) is placed at step 3 although it is detected by the validation in step 5; UC-08 A1 and A2 describe the same case; the pre-conditions of UC-12 and UC-14 require pending items although A1 handles an empty list; UC-14 never names the moderation actions; UC-07 and UC-10 have no exception for a failed image upload although Cloudinary is their secondary actor. (Found 11/11.)	UC-01, UC-07, UC-08, UC-10, UC-12, UC-14	**Fixed in this baseline.** UC-01 A1 moved to step 5. UC-08 A1 and A2 were merged. The pre-conditions of UC-12 and UC-14 only require a logged-in administrator. UC-14 names the actions Hide post, Remove post and No action required, as in the Glossary and the prototype. UC-07 E3 and UC-10 E3 cover a failed upload.
G. Prototype, traceability and text			
34	Section 8 (Prototype / Mockups) is empty in the draft of 10/11, although the Figma file "All Screen" contains screens for all 18 use cases. (Found 11/11.)	Section 8 UC-01 to UC-18	**Fixed in this baseline.** Section 8 lists the screens of every use case (124 frames: 62 desktop and 62 mobile) with links to the Figma pages and to the interactive prototype.

35	The RTM did not match the FR table: the FR-05 row pointed to UC-16 and UC-17 while FR-05 had become the history requirement; the FR-06 row said "Volunteer Task Board" while the Use Case List assigned UC-18 to FR-06; FR-07 had no row; UC-04 was traced to FR-03 in the list but to FR-03 and FR-04 in its specification; the Screen column used generic names instead of screen IDs. (Wrong FR mappings first found 03/11; found again 11/11.)	Section 9 FR-05, FR-06, FR-07 UC-04, UC-16, UC-17, UC-18	**Fixed in this baseline.** The RTM was rebuilt with one row per sub-requirement, the related use cases and the Figma screen IDs (SCR-xx_UC-xx). Should-have and Could-have requirements are listed with "no use case in this baseline".
36	The prototype shows content that the SRS does not specify: campaign search, category filter, sorting and "Load more" on the campaign list (G1 mentions "search", but no FR or use case step does); "Share campaign", an "Updates" tab and "How funds will be used" on the campaign page; organizer updates, expense reports and a 7-day chart on the progress page (F06 is Should-have); a required "Short summary" field in the campaign form (not in BR-02); the password rule "at least 8 characters, including a number"; acceptance of the Terms of use; a "View" link on each donation row. (Found 08/11 and 11/11.)	G1 BR-02 F06 UC-01, UC-04, UC-10, UC-17, UC-18	Fixed in this baseline. The Figma prototype was aligned with the SRS. Unsupported interactive features were removed or simplified to remain within the agreed PA1 scope. Password validation was formally specified in BR-19, UC-01, and the corresponding acceptance criteria. Retained interface elements were checked against the relevant requirements and use case specifications. No additional out-of-scope functionality was introduced.

37	The organizer and administrator campaign screens (UC-10 to UC-13) exist only as desktop frames, while OE-02 sets a minimum supported width of 360 px. (Found 11/11.)	OE-02 NFR-06 UC-10 to UC-13	**Fixed in this baseline.** A 360 px mobile version of each of the eight screens of UC-10 to UC-13 was added on the Figma pages and linked in the interactive prototype (see the notes under the Section 8 screen table).
38	Text defects in Sections 2–3, 5.3 and 7.8 of the merged draft: the feature name “F04 Campaign and moderation” lost the word “approval”; reference 2 points to /pa/PA0 for PA1; G4 says “forum post or comment” and UC-08 says that interactions are subject to moderation, although FR-04, BR-05 and UC-14 only cover posts; the Glossary defines a registered user as someone who “has logged in” although UC-02 starts with a registered user who is not logged in; “closesfundraising”; the source of BR-10 has an unclosed bracket. (Found 11/11.)	Section 2.1 (scope table, references) Glossary G4 BR-10 UC-08	**Fixed in this baseline.** The texts were corrected: “F04 Campaign approval & moderation”, “/pa/PA1”, “reported forum post”, “a person who has an ImpactHub account”, and the two typing errors; the moderation sentence was removed from UC-08.

Review result and baseline. All resolutions marked “Fixed” in the table above are included in this version of the SRS. This version becomes the requirements baseline of ImpactHub for software design (PA3), test planning (PA4) and implementation (PA5).

The 3 items marked "Deferred to PA3" (items 10, 11, and 12) are not part of the baseline. They remain open requirement questions to be addressed during PA3 and recorded in the PA3 requirement change list (WBS 3.6.2). Any later changes to baselined requirements will be documented and traced in subsequent project phases.

11 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Trần Phương Thúy	ChatGPT (GPT-6)	Requirements Review (Sec. 10), acceptance criteria, prompt for the Figma prototype	Checked against FRs, use cases, WBS and review records. Added IDs, Given–When–Then and negative cases; rejected merged issues and out-of-scope scenarios.
2	Hồng Gia Bảo	Gemini 3 Flash	W6 status report corrections; FR/NFR requirement review for the baseline	Checked against peer-review checklists, team guidelines and PA1 sources. Wording aligned with repo conventions, in-charge metadata customised, verbose narrative rejected.
3	Trần Anh Khoa	ChatGPT	Actors and use cases, UC diagram, UC-10–13, RTM, review table	Compared with PA1, FR-03/04, BR-02–09 and the Figma screens. Fixed statuses and mappings; removed identity verification and extra actors.
4	Lê Nho Duy Phước	Claude (Opus 5.5 / Fable 5), Claude Code + Figma	Sec. 2–3, FL-05 and All Screen mockups, SRS consistency review, Sec. 10 record and baseline edits	Checked against PA0 and the submitted PA1. Screenshots, contrast audit, link and frame counts, and OOXML validation. Gateway replaced by the demo wallet; 2 wrong findings withdrawn.